Correction D.S :

Ex1 : (4 pts)

$$f(x) = \begin{cases} e^{-(x-\theta)} & \text{si } x \geq 2 \\ 0 & \text{si non} \end{cases}$$

1) f est une densité de probabilité.

si $\begin{cases} f(x) \geq 0 \quad \forall x \in \mathbb{R} \\ \int_{-\infty}^{+\infty} f(x)\,dx = 1 \end{cases}$

- $f(x) = e^{-(x-\theta)} \geq 0 \quad \forall x \geq 2.$

$= 0 \geq 0 \quad \forall x < 2$

- $\int_{-\infty}^{+\infty} f(x)\,dx = \int_{2}^{+\infty} e^{-(x-\theta)}\,dx = 1$

$\Rightarrow \left[-e^{-(x-\theta)}\right]_{2}^{+\infty} = 1.$

$\Rightarrow e^{-(2-\theta)} = 1 \Rightarrow e^{-2} e^{+\theta} = 1$

$\Rightarrow e^{\theta} = e^{2} \rightarrow \boxed{\theta = 2}$

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$$f(x) = \begin{cases} e^{-(x-2)} & \text{si } x \geq 2 \\ 0 & \text{sinon.} \end{cases}$$

(2)

2)

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

·) si $x < 2 \Rightarrow F(x) = 0$.

·) si $x \geq 2 \Rightarrow F(x) = \int_{2}^{x} f(t)\,dt = \int_{2}^{x} e^{-(t-2)}\,dt$

$$= \left[-e^{-(t-2)}\right]_{2}^{x}$$

$$= -e^{-(x-2)} + e^{-(2-2)}$$

$$= 1 - e^{-(x-2)}$$

$$F(x) = \begin{cases} 1 - e^{-(x-2)} & \text{si } x \geq 2 \\ 0 & \text{sinon} \end{cases}$$

3)

$$P(-1 \leq X \leq 5) = P(X \leq 5) - P(X \leq -1)$$

$$= F(5) - F(-1)$$

$$= 1 - e^{-(5-2)} - 0$$

$$= 1 - e^{-3}$$

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4) $X \sim \mathcal{U}[-2,1]$

$$E[X] = \frac{-2+1}{2} = \frac{-1}{2}.$$

$$V[X] = \frac{(1+2)^2}{12} = \frac{3^2}{12} = \frac{9}{12} = \frac{3}{4}$$

<u>Ex 2</u> :

$$X \sim \mathcal{N}(1; 0{,}03) \rightsquigarrow Z = \frac{X-1}{0{,}03} \sim \mathcal{N}(0,1)$$

1) $$P(X > 1{,}05) = P\left(\frac{X-1}{0{,}03} > \frac{1{,}05-1}{0{,}03}\right)$$

$$= P(Z > 1{,}66) = 0{,}048.$$

2) $$P(0{,}94 \leq X \leq 1{,}08) = P\left(\frac{0{,}94-1}{0{,}03} \leq Z \leq \frac{1{,}08-1}{0{,}03}\right)$$

$$= P(-2 \leq Z \leq 2{,}66)$$

$$= P(Z \leq 2{,}66) - P(Z \leq -2)$$

$$= 1 - P(Z \geq 2{,}66) - P(Z \geq 2)$$

$$= 1 - 0{,}0039 - 0{,}022.$$

$$= 0{,}974.$$

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3) (4)

$$P(1-a \leq X \leq 1+a) = 0,2.$$

$$P(-a \leq X-1 \leq a) = 0,2.$$

$$P\left(\frac{-a}{0,03} \leq \frac{X-1}{0,03} \leq \frac{a}{0,03}\right) = 0,2$$

$$P\left(\frac{-a}{0,03} \leq Z \leq \frac{a}{0,03}\right) = 0,2$$

$$P\left(Z \leq \frac{a}{0,03}\right) - P\left(Z \leq -\frac{a}{0,03}\right) = 0,2.$$

$$1 - P\left(Z \geq \frac{a}{0,03}\right) - P\left(Z \geq \frac{a}{0,03}\right) = 0,2.$$

$$1 - 2P\left(Z \geq \frac{a}{0,03}\right) = 0,2$$

$$2P\left(Z \geq \frac{a}{0,03}\right) = 0,8$$

$$P\left(Z \geq \frac{a}{0,03}\right) = 0,4$$

$$\frac{a}{0,03} = 0,25 \longrightarrow a = 0,25 \times 0,03$$
$$= 0,00075.$$

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Ex3 :

$$f_\theta(x) = \begin{cases} \theta x^{-(\theta+1)} & \text{si } x > 1 \\ 0 & \text{sinon} \end{cases} \qquad \theta > 1$$

1) a) $\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_\theta(x)\, dx = \int_1^{+\infty} \theta x x^{-(\theta+1)} dx$

$$= \theta \int_1^{+\infty} x^{-\theta}\, dx = \left[\frac{\theta}{-\theta+1} x^{-\theta+1} \right]_1^{+\infty}$$

$$= \frac{-\theta}{-\theta+1} = \frac{\theta}{\theta-1}$$

b) $\overline{X}_n = \mathbb{E}[X] = \phi(\theta)$. avec $\phi(x) = \frac{x}{x-1}$

$$T_1 = \phi^{-1}(\overline{X}_n) \rightarrow \phi(T_1) = \overline{X}_n$$

$$\overline{X}_n = \frac{T_1}{T_1 - 1} \Rightarrow \overline{X}_n (T_1 - 1) = T_1.$$

$$\overline{X}_n T_1 - T_1 = \overline{X}_n$$

$$\Rightarrow T_1 (\overline{X}_n - 1) = \overline{X}_n$$

$$\Rightarrow \boxed{T_1 = \frac{\overline{X}_n}{\overline{X}_n - 1}}$$

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2) a) $\mathcal{L}(x_1, \dots, x_n, \theta) = \prod_{i=1}^{n} f_\theta(x_i) = \theta^n \prod_{i=1}^{n} x_i^{-(\theta+1)}$ (6)

b) $\ln \mathcal{L}(x_1, \dots, x_n, \theta) = \sum_{i=1}^{n} \ln\left(\theta\, x_i^{-(\theta+1)}\right)$

$$= n \ln(\theta) + \sum_{i=1}^{n} \ln\left(x_i^{-(\theta+1)}\right)$$

$$= n \ln(\theta) - (\theta+1) \sum_{i=1}^{n} \ln(x_i).$$

c) $\begin{cases} \frac{\partial}{\partial\theta} \ln \mathcal{L} = \frac{n}{\theta} - \sum_{i=1}^{n} \ln(x_i). \\ \frac{\partial^2}{\partial\theta^2} \ln \mathcal{L} = \frac{-n}{\theta^2} < 0 \end{cases}$ ✓

$$\frac{\partial}{\partial\theta} \ln \mathcal{L} = 0 \Rightarrow \frac{n}{\theta} - \sum_{i=1}^{n} \ln(x_i) = 0$$

$$\Rightarrow \frac{n}{\theta} = \sum_{i=1}^{n} \ln(x_i).$$

$$\Rightarrow \frac{1}{\theta} = \frac{1}{n} \sum_{i=1}^{n} \ln(x_i)$$

$$\boxed{T_2 = \frac{1}{\frac{1}{n} \sum_{i=1}^{n} \ln(x_i)}}$$

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